

Class 16: Intro to UNIX

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Basic Unix

Some important file system commands include: pwd: print working directory ls: list files and folders cd: change directory mkdir: new directory rm: delete files and directories nano: a very basic text editor touch: available less: to view/read text files page by page (page viewer) My AWS instance: ssh -i "Jas.pem" ubuntu@ec2-44-251-148-96.us-west-2.compute.amazonaws.com

Q. What does the star character accomplish here? Ask Barry, or your class neighbor, if you are not sure!

The star acts as a wild card.

Q. How many sequences are in this mouse.1.protein.faa file? Hint: Try using grep to figure this out...

there are 55052 sequences

Q. What happens if you run the above command without the > mm-first.fa part?

It will return the data but not make a file.

Q. What happens if you were to use two '>' symbols (i.e. » mm-first.fa)?

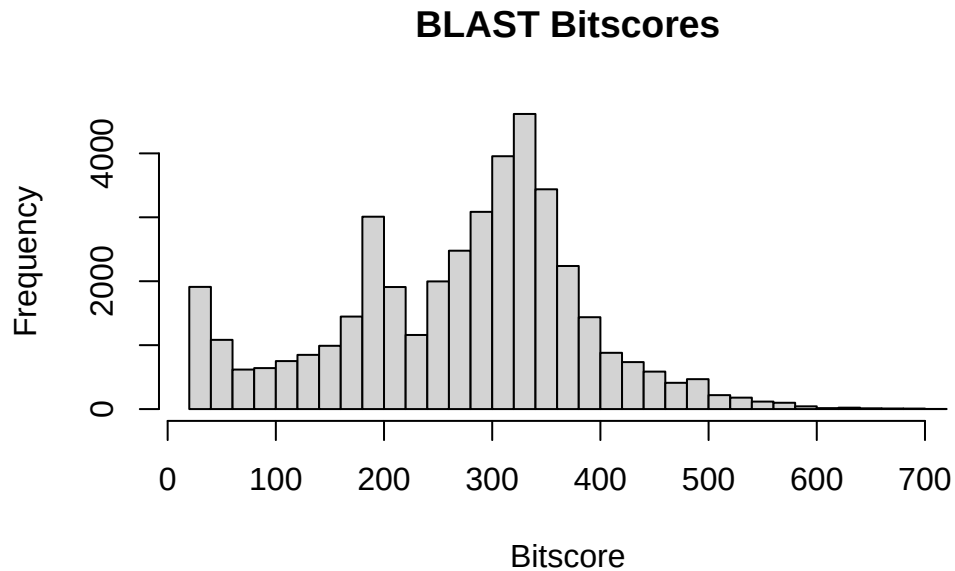
If you have a file with the name of the data you're using the "»" it will put the head data, if you don't have a file of that name it will do nothing.

Zebrafish results.

```
Zbruh <- read.delim("mm-second.x.zebrafish.tsv", header = FALSE)
colnames(Zbruh) <- c("qseqid", "sseqid", "pid", "length", "mismatch",
                    "gapopen", "qstart", "qend", "sstart", "send", "eval",
                    "bitscore")
```

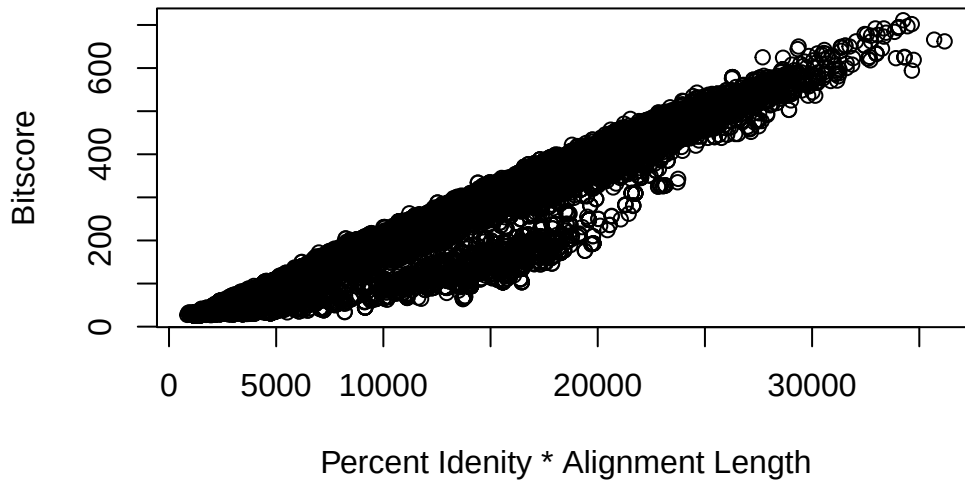
Making a histogram with R of bitscores

```
hist(Zbruch$bitscore,  
     breaks=30,  
     main = "BLAST Bitscores",  
     xlab = "Bitscore")
```



```
plot(Zbruch$pident * (Zbruch$qend - Zbruch$qstart),  
     Zbruch$bitscore,  
     xlab = "Percent Identity * Alignment Length",  
     ylab = "Bitscore",  
     main = "Relationship Between Identity and Bitscore")
```

Relationship Between Identity and Bitscore



Q. Note the addition of the `-r` option here: What is its purpose? Also what about the `*`, what is its purpose here?

The `-r` will copy directories instead of individual files. The `*` will be so that every file in the “work” directory is copied.